

ROC $\Rightarrow$ Imbalance $\rightarrow$ distributed

① Threshold modify: $\rightarrow$

TPR vs FPR

probability

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

dataset $\rightarrow$ Binary Classifier

1 $\rightarrow$ P, 0 $\rightarrow$ N

Actual y
 $y = 0, y_p \leq \text{Ther.}$
 $= 1, y_p > \text{Ther.}$

y_{pro}	$y(0)$	$y(0.2)$
0.8	1 TP	1 TP
0.96	1 FP	1 FP
0.4	1 TP	1 TP
0.3	1 TP	1 TP
0.2	1 FP	0 FN
0.7	1 TP	1 TP

$y(0.4)$	$y(0.6)$	$y(0.8)$
1 TP	1 TP	0 FN
1 FP	1 FP	1 FP
0 FN	0	0 FN
0 FN	0	0 FN
0 TN	0	0 TN
1 TP	1	0 FN

- ① $y(0)$, $TPR = \frac{4}{4+0} = 1$, $FPR = \frac{2}{2+0} = 1$
- ② $y(0.2)$, $TPR = \frac{4}{4+0} = 1$, $FPR = \frac{1}{1+1} = 0.5$
- ③ $y(0.4)$, $TPR = \frac{2}{2+2} = 0.5$, $FPR = \frac{1}{1+1} = 0.5$
- ④ $y(0.6)$, $TPR = \frac{2}{2+2} = 0.5$, $FPR = \frac{1}{1+1} = 0.5$
- ⑤ $y(0.8)$, $TPR = \frac{0}{4} = 0$, $FPR = 0.5$

TPR, FPR

TPR	FPR	Formula
1	1	$TPR = \frac{TP}{TP+FN}$
1	0.5	
0.5	0.5	$FPR = \frac{FP}{FP+TN}$
0.5	0.5	
0	0.5	

at dif.

Thresholds

ROC: TPR vs FPR at diff. Thresholds!

